PYTHON CASE STUDIES — EASY EXPLANATION

Quick one-page notes for Case Studies 3–6

3. Retail Billing with Multiple Discount Rules

Goal: Calculate a shop bill for many products, then apply category tax and discounts. A discount may depend on the bill amount, and the final discount must not exceed the allowed limit.

Read product names, prices and quantities → calculate each price × quantity → add to subtotal → add the correct tax → check discount conditions → apply the maximum discount limit → print invoice.

Use: lists to store products, loops to process each product, conditions to choose tax/discount rules, and operators for totals and percentages.

4. ATM Transaction Sequence Simulation

Goal: Simulate ATM transactions such as deposit, withdrawal or balance check while keeping the balance correct.

Start with the account balance → read each transaction → check transaction type and amount → if deposit, add money; if withdrawal, check sufficient funds and limits before subtracting → record success/failure → show the final balance and transaction summary.

Use: conditions for transaction rules, loops for multiple transactions, lists to keep transaction records, and variables to update the balance.

5. Password Policy Validation Engine

Goal: Decide whether a password follows the required security rules.

Read password → check minimum/maximum length → check required character types (uppercase, lowercase, digit, symbol) → check repeated characters if required → reject prohibited words/text → print valid or invalid and explain why.

Use: string operations to inspect characters, loops to examine each character, and conditions to test every rule. The exact rules come from the question.

6. Advanced Numeric Pattern Generator

Goal: Print a number pattern, such as a hollow triangle, pyramid or symmetric shape, for a given size.

Read the size → validate that it is positive → outer loop controls each row → inner loop decides what to print in each column → print numbers/spaces according to the pattern rule → move to the next line.

Use: nested loops (a loop inside another loop), conditions to decide borders/spaces, and print formatting for alignment.

Quick revision: Which Python concept does what?

Concept	Easy meaning	Examples here
Variables & operators	Store values and calculate	bill, balance, subtotal, percentages
Conditions (if/elif/else)	Make decisions based on rules	discounts, ATM limits, password checks
Loops & lists	Repeat steps and store many items	products, transactions, characters
Nested loops	Use one loop inside another	rows and columns of a pattern

Exam tip: Identify the inputs, write the steps in order, apply every condition carefully, update totals/balance only when allowed, and display a clear result. Use the exact values and rules stated in the question.